Parthib Saha

Frontend Engineer

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Summary

Frontend engineer building fast, tested React + TypeScript interfaces for real-time and AI-powered products: a collaborative code editor with live cursors and offline sync, a live e-commerce storefront, and a browser-based interview tool. Builds on reusable component libraries and backs the UI with **200+ frontend tests**.

Technical Skills

Core:	TypeScript, JavaScript (ES6+), React 19, HTML5, CSS3, responsive design, accessibility (ARIA)
State & Data:	Redux Toolkit, React Router 7, React Hook Form + Zod, Axios, REST, Socket.IO client
UI & Styling:	Tailwind CSS v4, Radix UI, Material UI, Framer Motion, Monaco Editor, Three.js, SVG charts
Real-Time & Browser:	Yjs CRDTs, WebSockets, WebRTC, IndexedDB, Web Speech API, FaceDetector API
Testing & Tooling:	Vitest, React Testing Library, Playwright, Vite, ESLint, pnpm, Git, GitHub Actions, Docker
Also:	Node.js, Express, FastAPI, PostgreSQL, MongoDB, Redis • C++, Python, SQL

Projects

DevArena – Real-Time Competitive Programming Platform **2025 – 2026**

React 19, TypeScript, Vite, Redux Toolkit, Tailwind CSS, Radix UI, Monaco, Yjs, Socket.IO, WebRTC, Vitest Source Code

- Built a **VS Code-style collaborative editor**: Monaco bound to **Yjs CRDTs** with **live remote cursors and selections**, presence, resizable panels, and undo/redo that reverts only your own edits.
- Made collaboration **work offline**: edits persist to **IndexedDB** and resync on reconnect with exponential backoff, fetching only the missed updates instead of the whole document.
- Built the 1v1 battle UI with **optimistic code submissions** reconciled against live Socket.IO verdicts, a battle timeline, replays, and WebRTC voice chat.
- Structured a **27k-line feature-based TypeScript app** on a reusable component library (Radix + Tailwind with typed variants), with **route-level code splitting**, **virtualized lists and tables**, hand-built SVG charts, error boundaries, and reduced-motion support.
- Wrote **208 frontend tests** (Vitest + React Testing Library) plus Playwright E2E, all run in GitHub Actions CI.

Smart Substitution & Product Recommendation Engine **2025 – 2026**

React 19, Vite, Material UI, React Router 7, React Hook Form, Zod, Stripe.js, Framer Motion Live Demo • Source Code

- Built the **live storefront**: searchable catalog, product pages showing the **top-5 AI-ranked in-stock substitutes** with a Gemini-written reason for each, cart, wishlist, and **Stripe checkout** with success and cancel flows.
- Built the full **auth experience**: login, signup, Google OAuth, password reset, a password-strength meter, **protected and admin-only routes**, and forms validated with **React Hook Form + Zod**.
- Backed the recommendation panel with a cached API that **cut redundant Gemini calls by ∼70%** and made **repeat loads near-instant**, with a fallback message so the panel never renders empty.

Multimodal AI Technical Interview Platform **2026**

JavaScript, Vite, Monaco Editor, Web Speech API, FaceDetector API, FastAPI, Gemini Source Code

- Built a **framework-free JavaScript** interview UI: resizable Monaco editor, **voice explanations via the Web Speech API**, and a **draggable picture-in-picture webcam** built on pointer events.
- Added **in-browser proctoring**: face detection with the FaceDetector API (plus a canvas fallback), tab-switch and paste tracking, a start-up grace period, and escalating severity levels.

Education

Birla Institute of Technology, Mesra – B.Tech in Artificial Intelligence **2023 – 2027 (expected)** • CGPA: **8.1 / 10**

Vivekananda Academy, Serampore – CBSE Class XII (2022): **88.8%** • Gospel Home School, Rishra – ICSE Class X (2020): **89.0%**

Achievements & Leadership

- **Semi-Finalist, Flipkart GRiD** (national SWE challenge); **Stage-2 qualifier, Maruti Suzuki XCELerate 2026**.
- Solved **400+ DSA problems in C++**, including **200+** on LeetCode; team entry at the **Adobe University Hackathon**.
- **Coordinator, NSS, BIT Mesra** – planned and ran volunteer drives and community outreach; **Technical Member, IEEE Student Branch** – helped organise technical events and workshops.